

Cafe POS App - Complete User Guide & Documentation

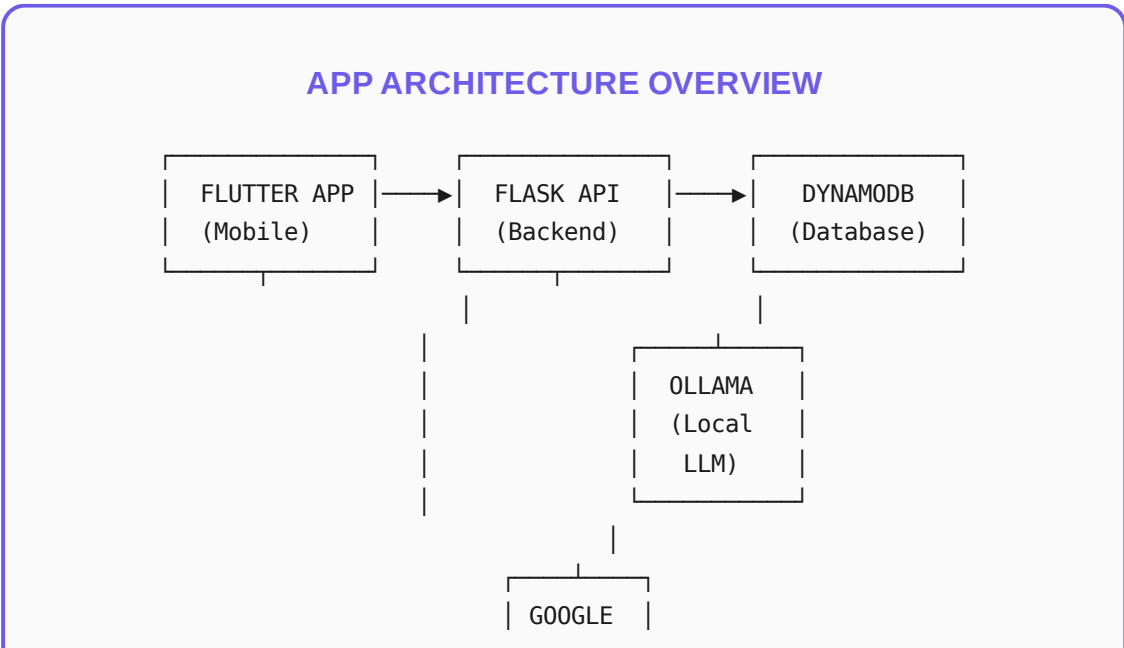
Version: 1.0 | **Platform:** Android & iOS | **Backend:** Flask + DynamoDB + AI
Currency: INR (₹) | **Last Updated:** September 2026

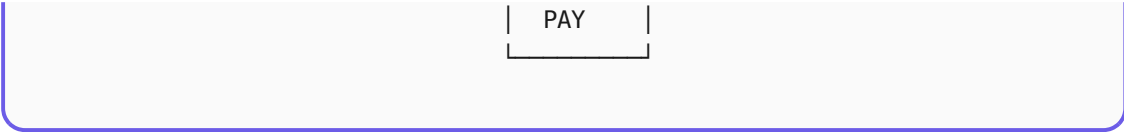
1. App Overview

The Cafe POS is an **internal point-of-sale app** designed for cafe staff and owners. It streamlines order taking, kitchen workflow, table management, billing, inventory, and vendor communication — all with built-in AI assistance.

Key Technology

- **Frontend:** Flutter (Android/iOS)
- **Backend:** Python Flask REST API
- **Database:** AWS DynamoDB
- **AI:** Local LLM via Ollama (llama3.2:3b)
- **Payments:** Google Pay, Cash, Card, UPI
- **Notifications:** Flutter Local Notifications





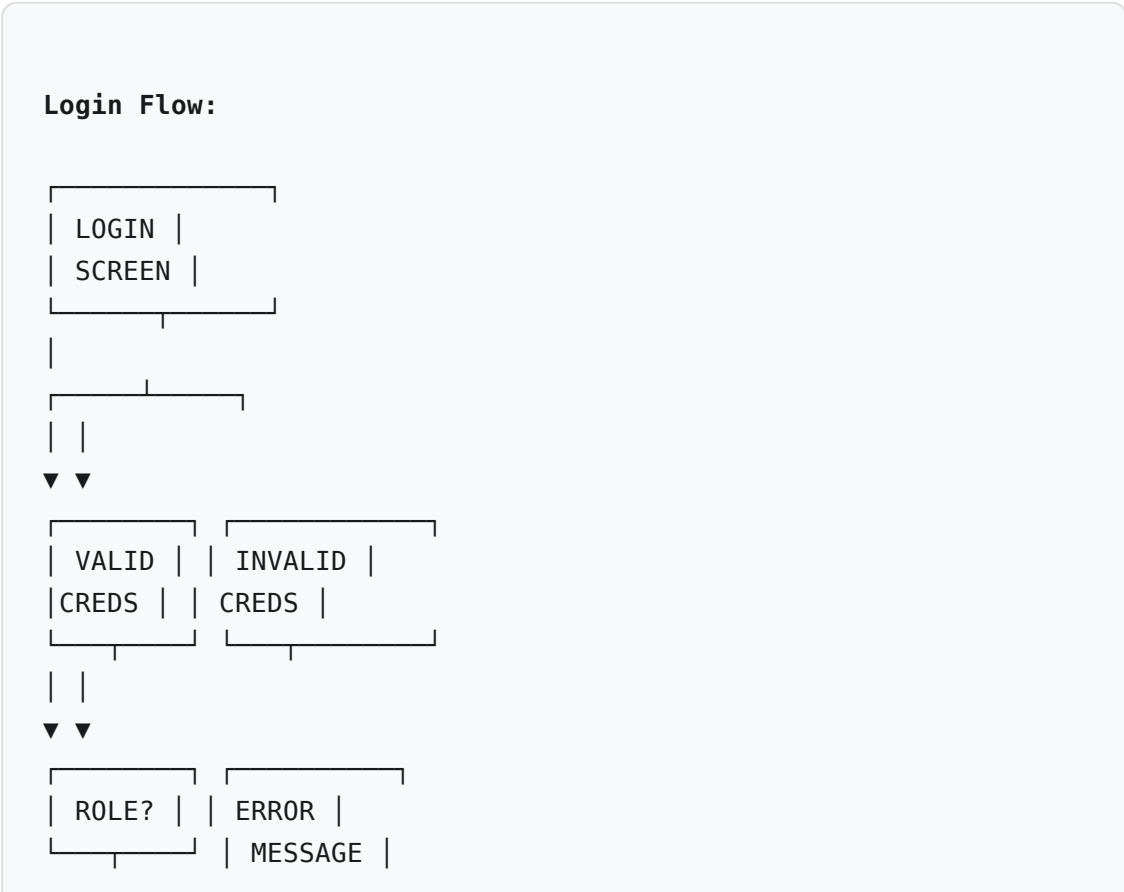
2. Login & Roles

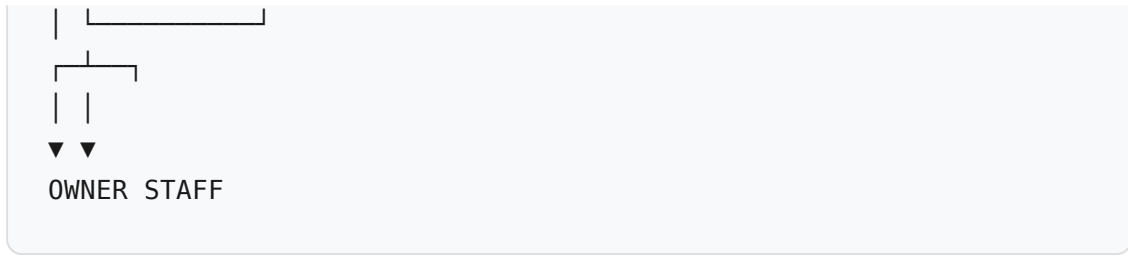
Default Credentials

Role	Username	Password	Access
Owner	owner	cafe123	Full access — all screens, settings, reports
Staff	priya	staff123	Limited — take orders, tables, order queue

First-Time Setup

1. Open the app → see the branded login screen with gradient background
2. Enter username and password → tap **Login**
3. Owner sees 5-tab dashboard | Staff sees 3-tab interface
4. **New staff?** Tap "Request Account" → fill name/username/password → owner must approve from Staff screen

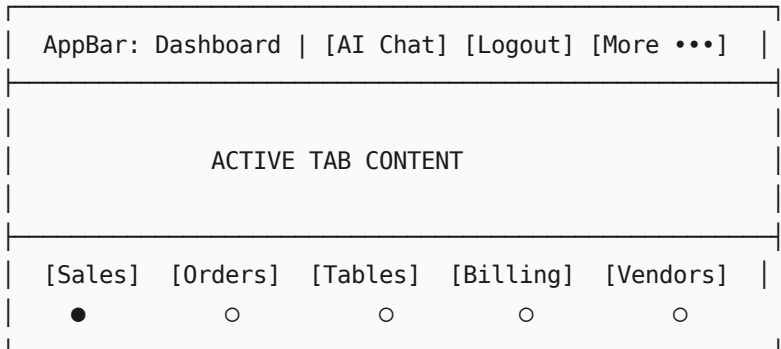




3. Navigation Structure

Owner Navigation (5 Tabs + More Menu)

OWNER SHELL

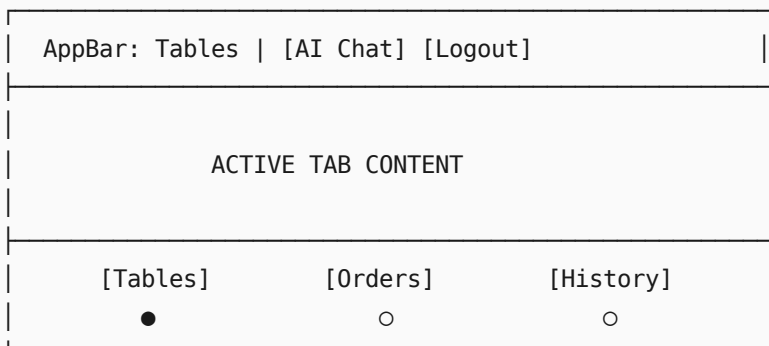


More Menu (... button):

- └ Reports → Daily sales + Monthly P&L
- └ Staff → Approve/remove team members
- └ Menu → Edit menu items & prices
- └ Inventory → Stock tracking & restock

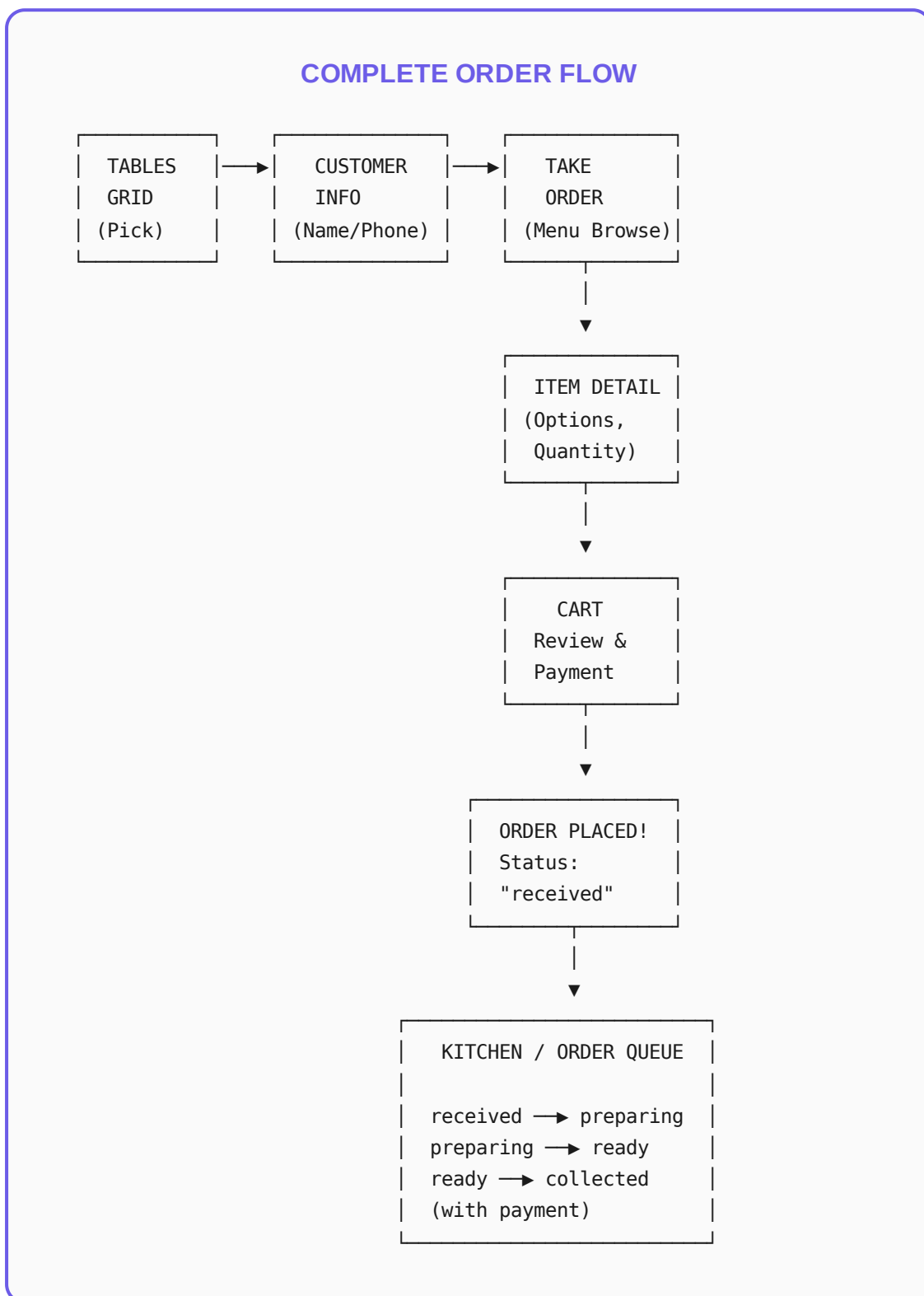
Staff Navigation (3 Tabs)

STAFF SHELL



4. Taking an Order (Staff Workflow)

This is the core workflow of the app. A staff member takes a customer's order, the kitchen prepares it, and the staff serves it to the customer.

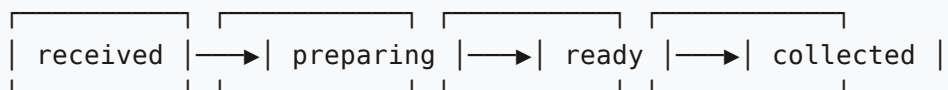


Step-by-Step: Taking an Order

Step	Screen	Action	Details
1	Tables Grid	Tap an empty (green) table	Table numbers shown. Green = available, Red = occupied.
2	Customer Info	Enter customer name (required), phone (optional)	Table badge shown. Tap "Start Order".
3	Take Order	Browse menu by category, tap items	Categories: Coffee, Tea, Food, Pastries. Sold-out items greyed out.
4	Item Detail	Select options (size, milk, etc.), set quantity, tap "Add to Cart"	Option groups: required/optional, single/multi-select.
5	Cart Review	Review items, adjust quantities, add notes, select payment method	AI "You might also like" suggestions shown. Payment: Cash/Card/UPI/GPay.
6	Place Order	Tap "Place Order"	Order created with status "received". Table marked occupied.

Order Lifecycle

Status Transitions:



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| | | |
| | | |
  
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Order placed Kitchen starts Food ready Payment received
 Kitchen sees working on it for pickup Order complete

NOTE: "collected" can be REOPENED back to "received" |
 Previously paid amount stays as credit. |

5. Owner Features

5.1 Dashboard (Sales Tab)

Sales Dashboard

Real-time sales analytics with beautiful charts:

- **Greeting:** "Good morning, Owner!" with current date
- **Date Picker:** Select any date to view sales
- **Range Selector:** Today / 7 Days / 30 Days
- **Total Sales Card:** Big number with trend indicator
- **Metrics:** Orders count, Average order value, Peak hour
- **Charts:** Sales trend bar chart, Payment method donut chart
- **Best Sellers:** Top items with quantity and revenue
- **Category Breakdown:** Sales by category (Coffee, Food, etc.)

5.2 Orders Queue

Orders Queue (Owner View)

See ALL orders across all staff members:

- **Status Summary Bar:** New | Preparing | Ready | Done counts
- **Active Toggle:** Show/hide completed orders
- **Order Cards:** Status badge, table number, customer name, phone, staff name, items, notes
- **Actions:** Add items, advance status, Reopen (for collected orders)
- **Tap card:** View full order details

5.3 Table Management

Tables Admin

- Grid view of all tables with status (empty/occupied)
- **Add Table:** FAB button → adds next available number
- **Tap occupied table:** See order details, items, advance or free
- **Free Table:** Clear table for next customer
- **Remove Table:** Delete unused table

5.4 Billing

Billing Screen

- Collected orders grouped by date
- **Summary:** Total Due / Paid / Unpaid amounts
- **Date Filter:** Pick specific date
- **Order Cards:** PAID/UNPAID chip, amount, customer
- **Mark as Paid:** Record payment received
- **Send Bill:** Opens WhatsApp with formatted bill + UPI link

5.5 Vendor Messaging

Vendor Message Screen

Easy restocking via WhatsApp:

- **Search:** Filter items by name
- **Sections:** Selected items, Low Stock, All Items
- **Quantity Controls:** +/- buttons for each item
- **Send via WhatsApp:** Opens WhatsApp with formatted order message
- **Mark Received:** Restocks inventory when delivery arrives

5.6 Reports (More Menu)

Reports Screen — Two Tabs

Daily Report:

- Date picker, total sales, order count
- Payment method breakdown (Cash/Card/UPI/GPay)
- Top selling items with quantities

Monthly P&L:

- Total Sales, Cost of Goods Sold, Net Profit/Loss
- Profit margin percentage
- Day-by-day breakdown
- AI-powered business suggestions

5.7 Staff Management (More Menu)

Staff Screen

- **Pending Approval:** New staff requests with Approve button
- **Active Team:** All approved staff with last login time
- **Owner Badge:** Shows who the owner is
- **Add Staff:** Create new staff directly (auto-approved)
- **Remove Staff:** Remove team member (with confirmation)

5.8 Menu Editor (More Menu)

Menu Editor

- List all menu items with images, names, categories, prices
- **Toggle Availability:** Switch on/off per item
- **Delete:** Remove menu item (with confirmation)
- **Add Item:** FAB → bottom sheet with Name, Category, Price, Description

5.9 Inventory (More Menu)

Inventory Management

- **Metric Cards:** Total Items count, Total Value (₹)
- **Item List:** Name, quantity, unit, cost price, low-stock badge
- **Add Item:** Name, quantity, unit dropdown (piece/kg/g/litre/ml/dozen/pack), cost, purchase date
- **Edit:** Modify item details
- **Restock:** Add quantity to existing item
- **Delete:** Remove item (with confirmation)
- **Low Stock Alert:** Items with ≤ 10 quantity highlighted in red

6. Staff Features

6.1 Tables Grid

Staff Tables View

- **Responsive Grid:** 3 columns (phone) / 4 columns (tablet)
- **Summary Bar:** Total | Occupied | Available counts
- **Green Cards:** Empty tables — tap to start order
- **Red Cards:** Occupied tables — tap to see order details
- **Table Locking:** Staff can only manage their own tables
- **Bottom Sheet:** Tap occupied table → see order, advance status, free table

6.2 Orders Queue

Staff Orders View

- **Auto-Refresh:** Updates every 3 seconds
- **Status Badges:** Color-coded (New=blue, Preparing=orange, Ready=green)
- **Active Toggle:** Show/hide completed orders
- **Order Cards:** Tap for full details, advance status, add items
- **Notifications:** Push notification when order reaches "ready"
- **Serve to Customer:** Green button on ready orders

6.3 Order History

Staff History

- Personal order history (filtered by logged-in staff)
- **Summary:** Orders taken, Total Revenue, Active count
- **Filter Chips:** All / Active / Completed
- **Order Cards:** Same design as order queue

6.4 Order Detail Screen

Order Detail

- **Status Badge:** Current order status with color
- **Who Took Order:** Staff name
- **When:** Date + time
- **Customer Info:** Name, phone, table number
- **Items List:** Each item with quantity, options, line total
- **Notes:** Special instructions
- **Payment Info:** Method, amount paid
- **"Serve to Customer" Button:** Green button when status=ready

6.5 AI Assistant

Ask Screen (AI Chat)

- **Suggestion Chips:** "Best sellers", "Busy tables", "Pending orders", "Free table"
- **Chat Interface:** Type questions, get AI-powered answers
- **Example Questions:**
 - "What are today's best sellers?"
 - "Which tables are busy?"
 - "How many orders are pending?"
 - "Free table 5"
- **Actionable Commands:** AI can propose actions (free table, advance order) that require your confirmation

7. Payment Methods

Method	How It Works	When Used
Cash	Customer pays cash, staff records payment	Manual payment collection
Card	Customer pays by card (POS machine)	Card payment at counter
UPI	Customer scans QR code, transfers money	UPI apps (PhonePe, Paytm, etc.)
GPay	Native Google Pay integration, auto-verification	Quick tap-to-pay

Payment Flow



Mark as
Paid +
Collected

8. AI Features

8.1 Smart Recommendations

During order taking, the AI suggests items based on what's already in the cart:

- "Customers who ordered Coffee also bought Croissant"
- "Popular with this table's previous orders"
- "Try our new seasonal Latte"

How it works: Uses local LLM (Ollama) running on your server. 25-second timeout, 2-minute cache.

8.2 AI Assistant (Agentic Q&A)

Chat-based assistant that can answer questions and perform actions:

- **Read-only queries:** "What's selling well today?" → instant answer
- **Actionable queries:** "Free table 3" → proposes action → you confirm → executes
- **Example commands:**

Query	AI Response
"What are the best sellers today?"	Shows top 5 items with quantities and revenue
"Which tables are occupied?"	Lists occupied tables with order details
"How many orders are pending?"	Shows count by status (received/preparing/ready)
"Free table 5"	Proposes action → Confirm/Cancel → Frees table
"Advance order #123"	Proposes action → Confirm/Cancel → Advances status

9. Table Management

Table Status Flow

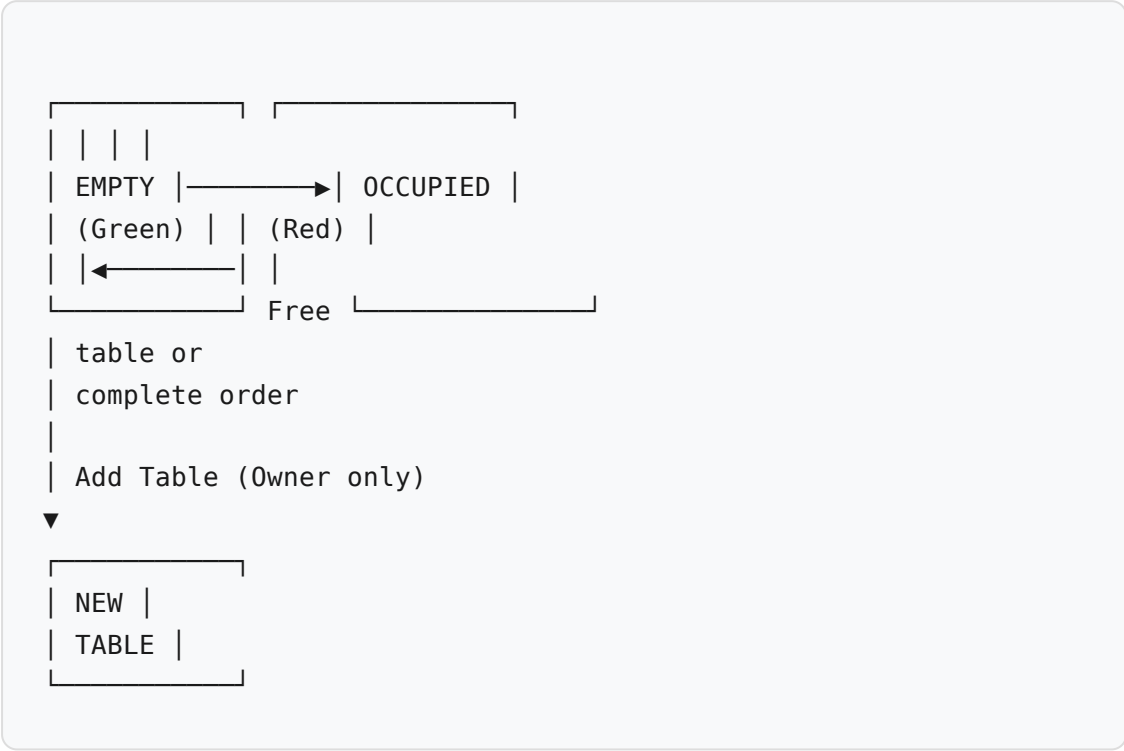


Table Locking

- When a staff member takes an order, they become the table's `takenBy`
- Other staff **cannot** advance or free that table
- **Owners can override** any table lock
- This prevents conflicts when multiple staff work simultaneously

10. Inventory Management

Inventory Units

Unit	Use Case
piece	Individual items (cups, napkins)
kg	Bulk items (coffee beans, sugar)
g	Small quantities (spices, tea leaves)

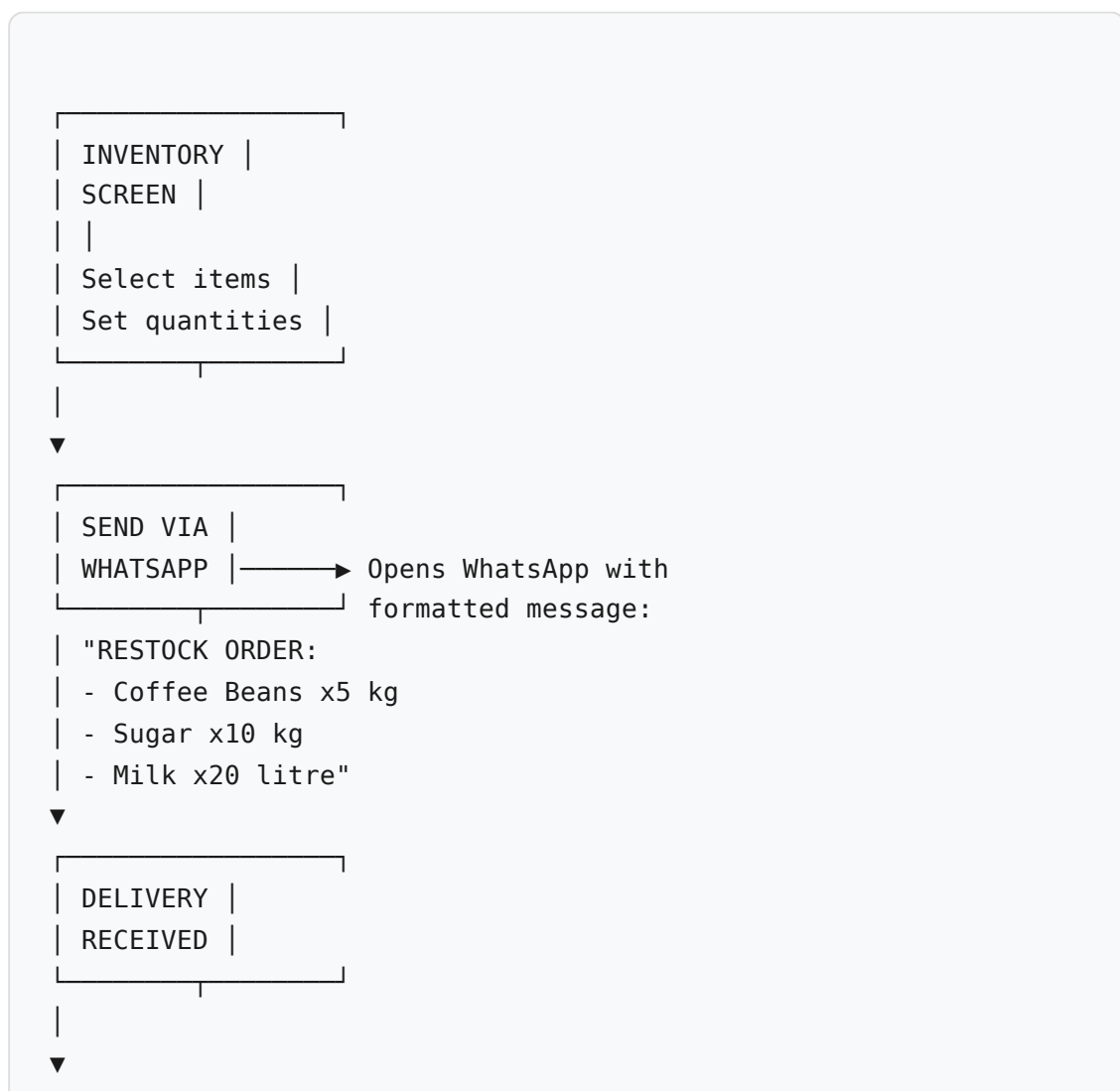
litre	Liquids (milk, syrups)
ml	Small liquid amounts (extracts)
dozen	Grouped items (pastries, eggs)
pack	Packaged items (filters, cups)

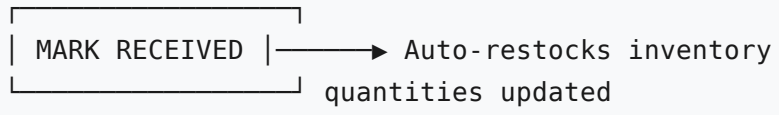
Low Stock Alerts

- Items with **≤10 quantity** are automatically flagged
- Shown with red "LOW" badge in inventory list
- Highlighted in Vendor Message screen for quick restocking

11. Vendor Communication

WhatsApp Restocking Flow





12. Setup & Installation

Prerequisites

- Flutter SDK installed
- Android Studio or Xcode
- AWS account (for DynamoDB + S3)
- Ollama installed (for AI features)

Backend Setup

1. Navigate to `backend/` directory
2. Install dependencies: `pip install -r requirements.txt`
3. Set environment variables:
 - `AWS_ACCESS_KEY_ID`
 - `AWS_SECRET_ACCESS_KEY`
 - `JWT_SECRET`
 - `OLLAMA_BASE_URL` (default: `http://localhost:11434`)
4. Run seed script: `python scripts/seed.py`
5. Start server: `python app.py`

Mobile App Setup

1. Navigate to `mobile/` directory
2. Install dependencies: `flutter pub get`
3. Update API URL in `lib/config.dart`
4. Run: `flutter run`

iOS IPA Build (GitHub Actions)

1. Push code to GitHub
2. Go to Actions tab → "Build iOS (Unsigned)"
3. Download IPA from Artifacts
4. Install via `ideviceinstaller` or TestFlight

13. API Endpoints Reference

Authentication

Method	Endpoint	Auth	Purpose
POST	/auth/login	None	Login, returns JWT token
POST	/auth/register	None	Register new staff (pending)

Menu

Method	Endpoint	Auth	Purpose
GET	/menu	None	List all menu items
GET	/menu/<id>	None	Get single item
GET	/categories	None	List categories

Orders

Method	Endpoint	Auth	Purpose
POST	/orders	Staff	Create new order
GET	/orders	Staff	List orders (filterable)
GET	/orders/<id>	Staff	Get order details
PATCH	/orders/<id>/status	Staff	Advance/set status
POST	/orders/<id>/reopen	Staff	Reopen collected order
POST	/orders/<id>/items	Staff	Add items to order

Tables

Method	Endpoint	Auth	Purpose
GET	/tables	Staff	List all tables
POST	/tables/<number>/free	Staff	Free a table

POST	/admin/tables	Owner	Add new table
DELETE	/admin/tables/<number>	Owner	Remove table

Admin (Owner Only)

Method	Endpoint	Purpose
GET	/admin/reports/daily	Daily sales report
GET	/admin/reports/range	Date range report
GET	/admin/reports/profit-loss	Monthly P&L
GET	/admin/billing	Billing summary
POST	/admin/billing/<id>/paid	Mark order paid
GET	/admin/staff	List staff
POST	/admin/staff	Create staff
POST	/admin/staff/<user>/approve	Approve staff
DELETE	/admin/staff/<user>	Remove staff
POST	/admin/menu	Create menu item
PUT	/admin/menu/<id>	Update menu item
DELETE	/admin/menu/<id>	Delete menu item
GET	/admin/inventory	List inventory
POST	/admin/inventory	Create inventory item
POST	/admin/inventory/<id>/restock	Restock item
DELETE	/admin/inventory/<id>	Delete inventory item

AI

Method	Endpoint	Auth	Purpose
POST	/recommend	Staff	AI cart suggestions
POST	/ask	Staff	Ask AI assistant

POST	/ask/execute	Staff	Execute AI-proposed action
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14. Quick Reference — Daily Operations

Morning Startup (Owner)

1. Open app → Login as owner
2. Check Dashboard for yesterday's sales
3. Check Inventory for low stock items
4. Use Vendor screen to send restock WhatsApp if needed
5. Review staff — approve any pending requests
6. Check Menu — update prices or mark items unavailable

During Service (Staff)

1. Open app → Login as staff
2. **Take Order:** Tables tab → tap green table → enter customer info → browse menu → add to cart → place order
3. **Check Status:** Orders tab → see queue with auto-refresh
4. **Serve Customer:** Tap "ready" order → "Serve to Customer"
5. **Add Items:** Tap order → "Add Items" → browse menu → add more
6. **Free Table:** Tables tab → tap occupied table → "Free Table"

End of Day (Owner)

1. Billing tab → check unpaid orders
2. Mark orders as paid (cash collected)
3. Send bills via WhatsApp for card/UPI orders
4. Reports tab → view daily summary
5. Check Inventory → restock any low items

Weekly (Owner)

1. Reports → Monthly P&L tab → check profit margin
2. Review AI suggestions for business improvements
3. Menu Editor → update seasonal items
4. Staff → check who's been logging in

5. Vendor → send bulk restock orders

Cafe POS App v1.0 — Built with Flutter + Flask + DynamoDB + AI